Supplementary Materials

Where Perinatal Behavioral Health Need Outpaces Treatment Capacity: An Open County-Level Need and Capacity Panel for the United States

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Contents: Table S1, state summary. Table S2, sensitivity analyses with Storey County values. File S3, codebook. File S4, capacity coding rules. File S5, quality assurance report. Full data, code, and provenance: https://github.com/memesobum/perinatal-bh-panel (Zenodo DOIs listed in the README).

Table S1. State summary

PPW SUD: substance use treatment listing with a pregnant/postpartum women program. HNLC: high need, low capacity.

State	Counties	Births	% counties no PPW SUD	% births in no-PPW counties	HNLC counties	PPW SUD listings per 10k births
AK	30	9,192	66.7	20.6	0	26.1
AL	67	58,019	64.2	31.1	7	7.1
AR	75	35,564	77.3	36	8	9
AZ	15	78,185	6.7	0.7	0	20.7
CA	58	413,272	13.8	0.6	1	8.9
CO	64	62,274	53.1	11.9	0	17.3
CT	9	34,609	0	0	0	19.6
DC	1	8,210	0	0	0	15.8
DE	3	10,575	0	0	0	23.6
FL	67	220,697	34.3	4.8	0	8.5
GA	159	125,109	64.8	20.6	2	9
HI	5	15,322	20	0	0	13.7
IA	99	36,469	56.6	28	0	18.6
ID	44	22,400	54.5	12.4	0	22.8
IL	102	128,454	64.7	16	0	11.8
IN	92	79,531	57.6	25.4	2	13.1
KS	105	34,395	72.4	26.7	0	16.9
KY	120	52,172	50	26.1	10	29.3
LA	64	56,284	54.7	24.1	4	13.5
MA	14	68,270	7.1	0.2	0	20.8
MD	24	67,555	12.5	1.4	0	24
ME	16	11,910	31.2	12.2	0	39.5
MI	83	102,140	39.8	10.3	3	13.7
MN	87	63,382	73.6	27.1	1	11.4
MO	115	68,521	73	35.3	4	8.8
MS	82	34,766	81.7	60.9	0	8.1
MT	56	11,158	80.4	30.5	1	17.9
NC	100	120,704	37	10.4	3	16
ND	53	9,779	84.9	29.1	1	13.3
NE	93	24,355	89.2	39.1	0	10.3
NH	10	12,213	20	8.6	0	21.3
NJ	21	101,797	4.8	0.7	0	15
NM	33	21,324	39.4	13.5	0	25.8
NV	17	32,888	58.8	2.5	0	12.8

State	Counties	Births	% counties no PPW SUD	% births in no-PPW counties	HNLC counties	PPW SUD listings per 10k births
NY	62	207,375	33.9	5.6	0	10.3
OH	88	128,307	37.5	12.5	1	16.8
OK	77	48,213	54.5	30.7	3	13.1
OR	36	39,567	30.6	7	1	20
PA	67	129,942	32.8	6.2	0	14.7
RI	5	10,180	20	3.3	0	15.7
SC	46	57,578	39.1	17.5	0	9
SD	66	11,255	87.9	46.8	4	10.7
TN	95	82,332	57.9	18.2	22	14.8
TX	254	383,771	78	14.3	21	4.5
UT	29	45,829	44.8	6.7	1	17.9
VA	133	94,707	47.4	14.9	6	14.6
VT	14	5,255	21.4	6.3	0	43.8
WA	39	82,725	41	5.3	0	9.7
WI	72	60,527	65.3	25.5	0	9.3
WV	55	16,913	50.9	28.3	1	31.9
WY	23	6,092	43.5	23.8	0	27.9

Table S2. Sensitivity analyses

Spearman rho: correlation of gap ranks with the baseline. Storey, NV gap illustrates small-county instability (blank when counties under 20 births are unranked).

Specification	Spearman rho	HNLC n	Overlap with baseline	Jaccard	Storey, NV gap
Baseline	1	107	107	1	-64.8
Capacity per 1,000 women aged 15-44	0.994	104	99	0.88	-63.5
Counties with <20 births unranked	1	110	101	0.87	nan
Health centers counted by address	1	106	104	0.95	-64.9
PPW programs counted by address	1	107	107	1	-64.8
Age-adjusted depression	0.989	115	99	0.8	-56.3
Need adds frequent mental distress	0.975	122	93	0.68	-66.3
Capacity adds MH listings	0.973	104	78	0.59	-51.3
Behavioral prescribers only	0.973	110	85	0.64	-65

File S3. Codebook

Plain-language summary: every column in every table, what it means, where it comes from, and how it was computed. One row in each county table is one U.S. county (or county-equivalent). Open this when a column name is unclear.

Key used in every county table: `fips` is the 5-digit county code (2-digit state + 3-digit county), as used by the Census Bureau in Vintage 2024 population estimates. Connecticut appears as its 9 planning regions (09110 to 09190), not its 8 former counties.

Universe: 3,144 counties and county-equivalents in the 50 states and the District of Columbia. Puerto Rico and the territories are not included in v0.1.

Missing values are blank cells. A blank is never a zero.

Release 1: `need_county.csv` (need measures)

Column	Definition	Source and year	How computed
fips	County code	Census PEP V2024	STATE + COUNTY
state_fips	State code	Census PEP V2024	

Column	Definition	Source and year	How computed
state	State postal abbreviation		from state_fips
county_name	County name	Census PEP V2024	CTYNAME
pop_total_2024	Total resident population, July 1, 2024	Census PEP V2024 `co-est2024-alldata`	POPESTIMATE2024
women_15_44	Female residents aged 15 to 44, July 1, 2024	Census PEP V2024 `cc-est2024-agesex-all`	AGE1544_FEM, YEAR = 6
women_15_44_year	Reference date of women_15_44		"2024-07-01"
births	Annual births used in the panel	see births_source	DECISION 1: NCHS 3-year average where available, otherwise Census PEP
births_source	Which births value was used		text
births_years	Years covered by `births`		"2021-2023" (NCHS) or "2023-07 to 2024-06" (Census)
births_nchs_3yr_avg	Average annual births to county residents, 2021 to 2023	AHRF 2024-2025, variable `births_3yr_avg_23` (NCHS Natality File)	as published; blank where NCHS suppressed it or for Connecticut planning regions
births_census_2024	Births, July 1, 2023 to June 30, 2024	Census PEP V2024 `co-est2024-alldata`, BIRTHS2024 (same values as AHRF `births_july_1_june_30_24`)	as published
general_fertility_rate	Births per 1,000 women aged 15 to 44	derived	1000 x births / women_15_44
depression_crude_pct	Percent of adults 18+ ever told they have a depressive disorder	CDC PLACES county data (model-based, from BRFSS)	measure DEPRESSION, crude prevalence
depression_crude_lcl, depression_crude_ucl	95% confidence limits for depression_crude_pct	CDC PLACES	
depression_ageadj_pct	Age-adjusted depression prevalence	CDC PLACES	measure DEPRESSION, age-adjusted
mental_distress_crude_pct	Percent of adults with 14+ days of poor mental health in past 30 days	CDC PLACES	measure MHLTH, crude
mental_distress_ageadj_pct	Age-adjusted version of the above	CDC PLACES	
places_release	PLACES release used		"PLACES 2025" (Socrata swc5-untb, BRFSS 2023); "PLACES 2024" (fu4u-a9bh, BRFSS 2022) only for Kentucky and Pennsylvania, which the 2025 release omits
places_brfss_year	BRFSS survey year behind the PLACES estimate	CDC PLACES	
births_x_depression	Births weighted by adult depression prevalence	derived	births x depression_crude_pct / 100. **A proxy for scale of need, not an estimate of perinatal depression cases.**
svi_overall	Overall Social Vulnerability Index percentile (0 = least, 1 = most vulnerable), ranked nationally	CDC/ATSDR SVI 2022, county, U.S. file	RPL_THEMES
svi_socioeconomic	SVI theme 1 percentile	CDC/ATSDR SVI 2022	RPL_THEME1
svi_household	SVI theme 2 percentile (household characteristics)	CDC/ATSDR SVI 2022	RPL_THEME2
svi_minority_language	SVI theme 3 percentile (racial and ethnic minority status)	CDC/ATSDR SVI 2022	RPL_THEME3
svi_housing_transport	SVI theme 4 percentile (housing type and transportation)	CDC/ATSDR SVI 2022	RPL_THEME4
svi_year	SVI vintage		"2022 (ACS 2018-2022)"
rucc_2023	Rural-Urban Continuum Code, 1 (largest metro) to 9 (most rural)	USDA ERS RUCC 2023	as published
rucc_year	RUCC vintage		"2023"
land_sqmi	Land area, square miles	Census 2024 Gazetteer, counties	ALAND_SQMI
pop_density_2024	Persons per square mile of	derived	pop_total_2024 / land_sqmi

Column	Definition	Source and year	How computed
	land		
rural	1 = nonmetro (RUCC 4 to 9), 0 = metro	derived	DECISION 2
frontier	1 = fewer than 6 persons per square mile	derived	DECISION 2
rurality_class	"frontier", "rural", or "metro"	derived	frontier takes precedence; 22 frontier counties are metro by RUCC (e.g. Matanuska-Susitna Borough, AK, RUCC 2)
flag_small_births	1 = births below 20	derived	rates and per-birth ratios are unstable (NCHS convention)
flag_births_census_fallback	1 = births came from Census, not NCHS	derived	
flag_places_prior_release	1 = depression from PLACES 2024 release	derived	Kentucky and Pennsylvania
flag_missing_depression	1 = no PLACES value	derived	Loving County, TX only
flag_missing_svi	1 = no SVI value	derived	

Release 2: capacity inventory

`capacity_facilities.csv` (one row per FindTreatment.gov listing)

Column	Definition
facility_id	Stable 12-character ID: first 12 hex digits of SHA-1 of lowercase name + street + ZIP + listing type. Used to remove exact duplicates.
location_id	12-character ID of lowercase street + ZIP. Listings sharing it are at one address.
query_state	State whose API query returned the listing
name, name2, street, city, state, zip, phone, website	As listed in FindTreatment.gov
latitude, longitude	As listed (WGS84). Blank for about 7% of listings.
facility_type	"SA" substance use, "MH" mental health, "OTP" opioid treatment program, as listed
serves_ppw	1 = "Special Programs/Groups Offered" (code SG) includes "Pregnant/postpartum women". **Only substance use listings carry this field.**
serves_adult_women	1 = SG includes "Adult women"
childcare_for_clients	1 = "Ancillary Services" (AS) includes "Child care for clients' children"
residential_beds_for_children	1 = AS includes "Residential beds for clients' children"
accepts_medicaid	1 = "Payment/Insurance/Funding Accepted" (PAY) includes "Medicaid"
offers_moud	1 = listing has an "Opioid Medications used in Treatment" (OM) entry
service_setting	Text of the "Service Setting" (SET) field
services_listed	0 = listing returned with no service detail (all OTP listings)
fips	County the listing is placed in
geo_method	"point_in_polygon" (coordinates inside a 2024 county boundary); "nearest_within_2km" (coordinates just outside every boundary, e.g. on a coastline; placed in the nearest county within 2 km; none occurred in v0.1); "zip_to_county" (no coordinates; ZIP rule in CODING_RULES.md); "zip_to_county_after_state_mismatch" (coordinates fell in a county of another state, so placed by ZIP instead); or "unassigned"

`capacity_hc_sites.csv` (one row per counted HRSA health center site)

Column	Definition
bphc_site_id	HRSA BPHC assigned site number
site_name, health_center_name, health_center_type	As published (FQHC or FQHC Look-Alike)
address, city, state, zip	Site address
fips	County code as published by HRSA
longitude, latitude	HRSA geocode
setting	Location setting description
operating_hours_per_week	As published
geo_method	"hrsa_county_code" (HRSA's own county code), "zip_to_county" (code missing; placed by ZIP), "unplaced_no_address_in_hrsa_file" (HRSA publishes no address), or "unassigned"
location_id	12-character ID of normalized address + ZIP; sites sharing it are

Column	Definition
	in one building

`capacity_clinicians_by_zip.csv` (**no names or NPIs**)

Column	Definition
zip5	Primary practice-location ZIP (5 digits)
fips	County the ZIP is assigned to (blank if unmatched)
group	obstetric, bh_prescriber, bh_therapist, or family_medicine (see CODING_RULES.md)
clinicians	Number of active individual NPIs

`capacity_county.csv`

Column	Definition
fips, state, county_name	Keys
tx_listings_all	All FindTreatment.gov listings in the county
tx_locations_all	Distinct addresses among those listings (one address often has both a mental health and a substance use listing)
tx_listings_mh, tx_listings_sa, tx_listings_otp	Listings by type
ppw_sud_facilities	Listings with serves_ppw = 1 (substance use treatment programs for pregnant/postpartum women)
ppw_sud_locations	Distinct addresses among those listings
women_program_facilities	Listings with serves_adult_women = 1
childcare_facilities, res_child_beds_facilities	Listings offering child care / residential beds for clients' children
ppw_sud_facilities_medicaid	ppw_sud listings that accept Medicaid
ppw_sud_facilities_moud	ppw_sud listings that offer medication for opioid use disorder
hc_sites	Counted HRSA health center sites
hc_locations	Distinct addresses among those sites
ob_clinicians	Individual NPIs, primary taxonomy in the obstetric group
bh_prescribers	Individual NPIs, primary taxonomy in the behavioral prescriber group
bh_therapists	Individual NPIs, primary taxonomy in the behavioral therapist group
bh_clinicians	bh_prescribers + bh_therapists
family_medicine_physicians	Individual NPIs with primary taxonomy 207Q00000X (reported separately)
*_per_1k_births	Count x 1,000 / births (births from need_county.csv). Blank where births = 0.
mh_listings_per_1k_births	Mental health listings per 1,000 births (not in the default gap score)
no_ppw_sud_facility	1 = ppw_sud_facilities is 0
no_bh_prescriber, no_ob_clinician	1 = count is 0
flag_small_count_ppw_sud	1 = 1 or 2 ppw_sud listings (one opening or closing changes the county's status)
findtreatment_access_date, nppes_file, hrsa_sites_file_date	Data dates for the capacity sources

Release 3: `gap_county.csv`

Column	Definition
fips, state, county_name, rurality_class, rural, frontier, births, flag_small_births	Carried from need table
ranked	1 = county is scored and ranked; 0 = left unranked because births are below MIN_BIRTHS_TO_RANK in code/config.py (0 in this build, so every county with births is ranked)
pr_depression_crude_pct, pr_svi_overall	National percentile rank (0 to 100) of each need component; higher = more need
need_score	Mean of the need percentile ranks
need_components_n	Number of need components available (2 except Loving County, TX)
pr_ppw_sud_facilities_per_1k_births, pr_bh_clinicians_per_1k_births, pr_ob_clinicians_per_1k_births, pr_hc_sites_per_1k_births	National percentile rank of each capacity component; higher = more capacity
capacity_score	Mean of the capacity percentile ranks
capacity_components_n	Number of capacity components available
gap_score	need_score minus capacity_score, -100 to +100. **Positive = need ranks above capacity.**
gap_rank_national	1 = largest gap in the U.S.
gap_rank_in_state	1 = largest gap in the state
high_need_low_capacity	1 = need_score in the top quarter AND capacity_score in the bottom quarter nationally
no_ppw_sud_facility, no_bh_prescriber, no_ob_clinician	Carried from capacity table

Percentile rank = 100 x (rank among ranked counties with a value / number of such counties), ties averaged. Ranks are relative: a county's score says where it stands against other counties, not whether services are adequate in absolute terms.

`panel_county.csv` joins all three county tables in one file for convenience.

File S4. Capacity coding rules

Plain-language summary: this file says exactly which facilities and clinicians are counted, and why. If you disagree with a rule, change it in `code/config.py` (or `code/04_build_capacity.py` for health centers) and re-run. Every rule here is a measurement decision Mr. Emesobum signs off on.

1. Treatment facilities (SAMHSA FindTreatment.gov)

How the listings were obtained. FindTreatment.gov has no single bulk file. The public locator API (Developer Guide v1.12, August 17, 2026) returns all listings in a state when called with `limitType=0` and the state's API ID (Appendix B of the guide; these IDs are not FIPS codes). The service type parameter `sType` was left out, so substance use, mental health, and opioid treatment program listings are all returned. 51 queries (50 states + DC), page size 2,000, accessed November 4, 2026. Every query returned its full reported count.

De-duplication. Listings with the same name, street, ZIP, and listing type are counted once (27,506 rows returned, 26,642 unique).

Placing listings in counties. Coordinates inside a 2024 Census county boundary (cartographic boundary file, 1:500,000). Listings with no coordinates are placed by ZIP code using the rule in section 4. **State check:** if the coordinates fall in a county of a different state than the listing's own state (a same-named town elsewhere, such as Arlington, OR placed in Texas), the listing is re-placed by ZIP, and only if the ZIP is in the listed state. 10 listings were corrected this way, 2 of them pregnant/postpartum programs. 85 listings (mostly P.O. box ZIP codes, plus Hartford, VT) could not be placed and are listed in the QA report.

Pregnant/postpartum flag (`serves_ppw`). The listing's "Special Programs/Groups Offered" field (code `SG`) contains the exact value "Pregnant/postpartum women". This is what the facility reported to SAMHSA's annual survey; it is not verified by this project.

Important: only substance use (SA) listings carry the pregnant/postpartum field. None of the 12,637 mental health listings and none of the 1,958 opioid treatment program listings have it. The facility measure is therefore **substance use treatment for pregnant and postpartum women**, named `ppw_sud_facilities`. Federally certified opioid treatment programs must give pregnant women priority admission, but they are counted separately (`tx_listings_otp`) and not as pregnant/postpartum programs.

Supporting flags (not in the gap score): "Adult women" (SG), "Child care for clients' children" (AS), "Residential beds for clients' children" (AS), Medicaid accepted (PAY), medications for opioid use disorder offered (OM).

2. Health center sites (HRSA)

From HRSA's "Health Center Service Delivery and Look-Alike Sites" file (downloaded November 4, 2026), a site is counted when all are true:

- Site status is Active.
- Site type is "Service Delivery Site" or "Administrative/Service Delivery Site" (administrative-only offices are not counted).
- Location type is "Permanent" (seasonal sites and mobile vans are not counted).
- Setting is not School, Correctional Facility, or Transitional Care in Carceral Setting (these do not serve the general population of pregnant women).

19,528 sites in the file; 11,992 meet the rules. County comes from HRSA's own county code; one site with a blank code (Oakland, CA) is placed by ZIP. Of the 11,992: 11,769 are in the 50 states and DC; 165 are in Puerto Rico and the territories; 58 have no address in HRSA's file (state shown as "XX") and cannot be placed. The 11,769 sites sit at 11,219 distinct addresses, because one building often holds several sites of the same health center (for example medical and dental). The gap score counts sites; `hc_locations` counts addresses.

3. Clinicians (CMS NPPES, November 2026 file)

- Individuals only (entity type 1). Organizations are not counted.
- Active: no deactivation date, or reactivated after deactivation.
- **Each clinician counted once**, in the group of their **primary taxonomy** (the code marked primary; if none is marked, the first listed code).
- **Location** = primary business practice address, U.S. only. Secondary practice locations are not counted.
- An NPI shows that a clinician registered to bill. It does not prove they are currently practicing, licensed in that state, seeing pregnant patients, or full-time.

Group	NUCC codes (v26.1)
Obstetric (`ob_clinicians`)	207V00000X OB/GYN; 207VX0000X Obstetrics; 207VM0101X Maternal & Fetal Medicine; 367A00000X Advanced Practice Midwife; 176B00000X Midwife; 363LX0001X NP Obstetrics & Gynecology; 363LW0102X NP Women's Health; 363LP1700X NP Perinatal; 364SW0102X CNS Women's Health; 364SP1700X CNS Perinatal
Behavioral prescriber (`bh_prescribers`)	2084P0800X Psychiatry; 2084P0802X Addiction Psychiatry; 2084P0015X Psychosomatic Medicine; 2084A0401X, 207QA0401X, 207RA0401X, 207LA0401X, 2083A0300X Addiction Medicine (all boards); 363LP0808X Psychiatric/Mental Health NP; 364SP0808X, 364SP0809X, 364SP0810X, 364SP0812X Psychiatric/Mental Health CNS
Behavioral therapist (`bh_therapists`)	Psychologists: 103T00000X, 103TA0400X, 103TB0200X, 103TC0700X, 103TC1900X, 103TF0000X, 103TH0004X, 103TH0100X, 103TP0016X, 103TP0814X, 103TP2700X, 103TP2701X, 103TW0100X; 1041C0700X Clinical Social Worker; Counselors 101Y00000X, 101YA0400X, 101YM0800X, 101YP2500X; 106H00000X Marriage & Family Therapist
Family medicine (reported only)	207Q00000X Family Medicine

Left out on purpose: gynecology-only and gynecologic subspecialties (207VG0400X, 207VX0201X, 207VE0102X, 207VF0040X, and others), lay midwives (175M00000X), child and adolescent psychiatry, school psychologists and counselors, pastoral counselors, social workers without the clinical specialty, and registered nurses. Family physicians are reported but not added to obstetric capacity, because NPPES cannot show which of them deliver babies.

4. ZIP code to county

- Each ZIP is treated as its ZIP Code Tabulation Area (ZCTA) and assigned to the county holding the largest share of its land area (Census 2020 ZCTA-to-county relationship file).
- **Connecticut:** that file still uses the old 8 counties. Connecticut ZCTAs are placed in the 2022 planning regions by their 2024 Gazetteer interior point.
- ZIPs that are not ZCTAs (many P.O. box and single-building ZIPs) are left unmatched: 12,439 clinicians (0.8%), listed by count in the QA report.

5. Small numbers

- `flag_small_births`: births below 20 (NCHS convention for unreliable rates). Per-birth capacity ratios in these counties swing widely; 2 clinicians in a county with 18 births looks like very high capacity.
- `flag_small_count_ppw_sud`: 1 or 2 pregnant/postpartum listings.
- Counties with 0 births have no per-birth ratios and no capacity score (Kalawao County, HI).

File S5. Quality assurance report (v0.1.0)

```
QA REPORT  U.S. County Perinatal Behavioral Health Need and Capacity Panel  v0.1.0
Plain-language summary: this file is the build's self-check. Each section says what was
counted and whether it passed. The Nevada section is for comparing with your county maps.

1. COUNTY UNIVERSE AND JOINS (need table)
  counties (Census PEP V2024, 50 states + DC): 3144  [expect 3144]  PASS
  duplicate FIPS: 0  PASS
  missing women_15_44: 0  []
  missing births: 0  []
  missing depression_crude_pct: 1  ['48301 Loving County']
  missing svi_overall: 0  []
  missing rucc_2023: 0  []
  missing land_sqmi: 0  []
  births source: NCHS 3-yr avg 3096; Census fallback 48 (9 Connecticut planning regions + 39 counties NCHS suppressed)
  PLACES prior release (2024, BRFSS 2022) used for: 187 counties (KY, PA)

2. RANGE AND SANITY CHECKS (need table)
  depression_crude_pct: min 12.30 median 23.00 max 34.30; outside [5,40]: 0  PASS
  svi_overall: min 0.00 median 0.50 max 1.00; outside [0,1]: 0  PASS
  general_fertility_rate: min 0.00 median 60.00 max 226.30; outside [10,150]: 8  CHECK
  rucc_2023: min 1.00 median 6.00 max 9.00; outside [1,9]: 0  PASS
    GFR outlier: 02158 Kusilvak Census Area, AK: births 224.0, women 15-44 1473
    GFR outlier: 02282 Yakutat City and Borough, AK: births 1.0, women 15-44 106
    GFR outlier: 15005 Kalawao County, HI: births 0.0, women 15-44 8
    GFR outlier: 31183 Wheeler County, NE: births 16.0, women 15-44 98
    GFR outlier: 46041 Dewey County, SD: births 181.0, women 15-44 963
    GFR outlier: 46095 Mellette County, SD: births 74.0, women 15-44 327
    GFR outlier: 48301 Loving County, TX: births 1.0, women 15-44 6
    GFR outlier: 51685 Manassas Park city, VA: births 33.0, women 15-44 3644
  births sum: 3,642,063  (U.S. registers about 3.6 million births a year)
  women 15-44 sum: 67,402,866

3. NEVADA NEED VALUES
      county_name rurality_class  rucc_2023  pop_density_2024  births  women_15_44  depression_crude_pct  svi_overall
      Carson City          metro          3            402.31   512.0         9966                  21.4       0.8508
 Churchill County       frontier          6              5.26   280.0         4666                  22.0       0.6271
     Clark County          metro          1            303.98 24629.0       486069                  20.3       0.8635
   Douglas County          rural          4             69.76   323.0         6516                  18.9       0.2059
      Elko County       frontier          5              3.17   676.0        10729                  21.5       0.6185
 Esmeralda County       frontier          9              0.20     2.0           91                  20.2       0.9077
    Eureka County       frontier          9              0.45    18.0          292                  23.5       0.4254
  Humboldt County       frontier          7              1.78   204.0         3233                  21.7       0.6160
    Lander County       frontier          9              1.05    61.0         1122                  22.8       0.5762
   Lincoln County       frontier          8              0.41    52.0          662                  21.2       0.5749
      Lyon County          metro          2             31.81   622.0        10713                  21.4       0.5593
   Mineral County       frontier          8              1.19    33.0          697                  22.3       0.5501
      Nye County       frontier          4              3.08   382.0         7653                  21.0       0.7954
  Pershing County       frontier          8              1.08    42.0          725                  19.6       0.6761
    Storey County          metro          2             15.59    18.0          536                  19.7       0.1836
    Washoe County          metro          2             80.32  4949.0       100957                  21.1       0.6971
White Pine County       frontier          9              0.96    85.0         1313                  21.5       0.2498

4. FINDTREATMENT.GOV LISTINGS
  rows returned by 51 state queries: 27,506
  unique listings after de-duplication (name+street+ZIP+type): 26,642
  unique street addresses (locations) among placed listings: 18,908  (one place often holds both a mental health and a substance use listing)
  placed in a county: 26,557 (99.7%)  methods: {'point_in_polygon': 24885, 'zip_to_county': 1662, 'unassigned': 85, 'zip_to_county_after_state_mismatch': 10}
  coordinates pointed to a county in another state, re-placed by ZIP: 10
  listing types: {'MH': 12637, 'SA': 12047, 'OTP': 1958}
  listings coded 'Pregnant/postpartum women': 4,600
  unassigned listings (85), mostly P.O. box ZIPs; first 10:
    Set Free Alaska | PO BOX 876741 | Wasilla, AK 99687
    Resilience Reach 907 | P.O. Box 876646 | Wasilla, AK 99687
    Meadows | 3400 Sabin Brown Road | Wickenburg, AZ 85358
    Meadows | 3400 Sabin Brown Road | Wickenburg, AZ 85358
    Intermountain Centers for | P.O. Box 86537 | Tucson, AZ 85754
    Intermountain Centers for | P.O. Box 86537 | Tucson, AZ 85754
    Intermountain Centers for Human Develo | P.O. Box 86537 | Tucson, AZ 85754
    SouthEastern AZ Behav Hlth Servs Inc | 1609 South 1st Avenue | Safford, AZ 85548
    Centers for Youth and Families | P.O. Box 251950 | Monticello, AR 71657
    Birch Tree Communities Inc | P.O. Box 1589 | Benton, AR 72018

5. HRSA HEALTH CENTER SITES
  active sites in file: 19,528; counted under coding rules: 11,992
  placement: {'hrsa_county_code': 11928, 'unplaced_no_address_in_hrsa_file': 58, 'unassigned': 5, 'zip_to_county': 1}
  in the 50 states + DC: 11,769 sites at 11,219 distinct addresses
  in Puerto Rico and territories: 160 with a territory county code, plus 5 in the Marshall Islands and Palau with none
  no address published by HRSA (state 'XX'), cannot be placed: 58

6. NPPES CLINICIANS
  NPPES records read: 9,891,116; individual (type 1): 7,540,866
  kept by primary taxonomy: 1,565,614; U.S. practice address: 1,564,724
  ZIP-to-county: {'zcta_largest_land': 1528728, 'ct_zcta_point_to_planning_region': 23557, 'unmatched': 12439}
  in territories (outside panel): 8,034
  by group: {'bh_therapist': 1198337, 'family_medicine': 157532, 'bh_prescriber': 124593, 'obstetric': 84262}
  External check vs AHRF (AMA Physician Masterfile 2023, patient-care MDs):
    obstetric clinicians vs AHRF OB/GYN: correlation of log counts r = 0.925 (n=3135)
    BH prescribers vs AHRF psychiatrists: r = 0.885 (n=3135)
    Note: NPPES group includes midwives and NPs; AHRF counts MDs only, so NPPES totals run higher.

7. NEVADA CAPACITY VALUES
      county_name  tx_listings_all  tx_locations_all  ppw_sud_facilities  hc_sites  ob_clinicians  bh_prescribers  bh_therapists  family_medicine_physicians
      Carson City               14                10                   4         1             17              19            229                          39
 Churchill County                5                 3                   1         0              4               3            101                          18
     Clark County              147               108                  23        38            414             673           6453                         867
   Douglas County                2                 2                   0         0              4               6             86                          24
      Elko County                3                 2                   1         4              7               7             70                           9
 Esmeralda County                0                 0                   0         0              0               0              0                           0
    Eureka County                0                 0                   0         0              0               0              0                           1
  Humboldt County                3                 2                   0         0              2               1             22                           3
    Lander County                1                 1                   0         1              0               1              3                           3
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Lincoln County	1	1	0	0	0	0	5	2
Lyon County	13	7	5	0	1	0	62	7
Mineral County	0	0	0	0	0	0	6	3
Nye County	12	9	2	3	3	4	94	16
Pershing County	1	1	0	0	0	1	5	7
Storey County	0	0	0	1	2	2	19	10
Washoe County	26	20	6	12	99	161	2043	248
White Pine County	1	1	0	0	0	1	36	4

Nevada listings coded Pregnant/postpartum women (42):

Churchill County	SA	New Frontier | 1490 Grimes Street, Fallon
Clark County	SA	ABC Therapy LLC | 7 Water Street, Henderson
Clark County	SA	ABC Therapy LLC | 730 North Eastern Avenue, Las Vegas
Clark County	SA	BHG Cheyenne Treatment Center | 3470 West Cheyenne Avenue, North Las Vegas
Clark County	SA	BHG Desert Inn Treatment Center | 3050 East Desert Inn Road, Las Vegas
Clark County	SA	BHG McDaniel Treatment Center | 2290 McDaniel Street, North Las Vegas
Clark County	SA	Better Choice Treatment Center | 198 East Ebb Tide Circle, Las Vegas
Clark County	SA	Care Clinic | 1050 East Flamingo Road, Las Vegas
Clark County	SA	Desert Parkway Behav Healthcare Hosp | 3247 South Maryland Parkway, Las Vegas
Clark County	SA	Desert Treatment Clinic LLC | 2598 Windmill Pkwy, Henderson
Clark County	SA	Dr Miriam/Sheldon G Adelson | 3661 South Maryland Parkway, Las Vegas
Clark County	SA	Henderson Comprehensive Treatment Ctr | 1536 North Boulder Highway, Henderson
Clark County	SA	Human Behavior Institute | nan, Las Vegas
Clark County	SA	Las Vegas Comprehensive Treatment Ctr | 2887 South Maryland Parkway, Las Vegas
Clark County	SA	Legacy Health and Wellness | 2921 North Tenaya Way, Las Vegas
Clark County	SA	Mind Body Solutions | 2725 South Jones Boulevard, Las Vegas
Clark County	SA	Multicultural Csl Collaborative Inc | 2920 South Jones Boulevard, Las Vegas
Clark County	SA	New Beginnings Counseling Centers Inc. | 3675 Pecos McLeod, Las Vegas
Clark County	SA	Partida Corona Medical Center | 2950 East Flamingo Road, Las Vegas
Clark County	SA	Southwest Vegas Comprehensive Trt Ctr | 6080 South Fort Apache Road, Las Vegas
Clark County	SA	THRIVE Solutions | 5584 South Fort Apache Road, Las Vegas
Clark County	SA	Vegas Treatment Center | 2121 East Flamingo Road, Las Vegas
Clark County	SA	Vogue Recovery Center | 4011 McLeod Drive, Las Vegas
Clark County	SA	WestCare Nevada Inc | 5659 Duncan Drive, Las Vegas
Elko County	SA	Vitality Integrated Programs Elko | 215 Bluffs Avenue, Elko
Lyon County	SA	Rural Nevada Counseling | 720 South Main Street, Yerington
Lyon County	SA	Rural Nevada Counseling | 801 Overland Loop, Dayton
Lyon County	SA	Rural Nevada Counseling | 3595 U.S. Highway 50, Silver Springs
Lyon County	SA	Rural Nevada Counseling | 415 U.S. Highway 95-A, Fernley
Lyon County	SA	Vitality Integrated Programs Dayton | 120 Pike Street, Dayton
Nye County	SA	Living Free Health and Fitness | 2050 NV Highway 160, Pahrump
Nye County	SA	New Beginnings Counseling Centers Inc | 2280 Calavada Boulevard, Pahrump
Washoe County	SA	BHG Reno Treatment Center | 160 Hubbard Way, Reno
Washoe County	SA	Bristlecone Family Resources | 704 Mill Street, Reno
Washoe County	SA	Empowerment Center | 7400 South Virginia Street, Reno
Washoe County	SA	Life Change Center | 1755 Sullivan Lane, Sparks
Washoe County	SA	Quest Counseling and Consulting | 6110 Plumas Street, Reno
Washoe County	SA	Vitality Integrated Programs | 1135 Terminal Way, Reno
Carson City	SA	Carson City Community Counseling Ctr | 788 Fairview Drive, Carson City
Carson City	SA	Carson City Community Counseling Ctr | 205 South Pratt Street, Carson City
Carson City	SA	Vitality Integrated Programs | 119 Long Street, Carson City
Carson City	SA	Vitality Unlimited | 900 East Long Street, Carson City

8. GAP SCORES
  minimum births to be ranked: 0 (0 = every county ranked); ranked counties: 3144
  need_score: min 1.00 median 51.20 max 96.80; outside [0,100]: 0 PASS
  capacity_score: min 16.00 median 50.40 max 99.70; outside [0,100]: 0 PASS
  gap_score: min -69.80 median -0.10 max 72.80; outside [-100,100]: 0 PASS

9. NEVADA GAP VALUES

county_name	births	need_score	capacity_score	gap_score	gap_rank_in_state
Carson City	512.0	58.8	79.4	-20.6	15.0
Churchill County	280.0	50.8	61.8	-11.0	12.0
Clark County	24629.0	53.2	62.0	-8.8	10.0
Douglas County	323.0	14.6	44.6	-30.0	16.0
Elko County	676.0	47.6	58.5	-10.9	11.0
Esmeralda County	2.0	54.8	16.0	38.8	1.0
Eureka County	18.0	49.2	16.0	33.2	2.0
Humboldt County	204.0	48.7	34.6	14.1	5.0
Lander County	61.0	52.7	40.1	12.6	7.0
Lincoln County	52.0	44.0	24.1	19.9	3.0
Lyon County	622.0	44.2	45.4	-1.2	8.0
Mineral County	33.0	48.4	29.3	19.1	4.0
Nye County	382.0	53.8	71.2	-17.4	13.0
Pershing County	42.0	40.5	27.0	13.5	6.0
Storey County	18.0	16.4	81.2	-64.8	17.0
Washoe County	4949.0	49.5	69.3	-19.8	14.0
White Pine County	85.0	29.2	36.8	-7.6	9.0

10. SPOT CHECKS (random seed 20261104)
  largest gap 27087 Mahnomen County, MN: births 77, ppw_sud 0, bh_clin 2, ob 0, hc 0, need 90.6, cap 17.8, gap 72.8
  largest gap 47033 Crockett County, TN: births 164, ppw_sud 0, bh_clin 2, ob 0, hc 0, need 82.2, cap 16.9, gap 65.3
  largest gap 47097 Lauderdale County, TN: births 285, ppw_sud 0, bh_clin 9, ob 0, hc 1, need 92.5, cap 28.2, gap 64.3
  largest gap 40067 Jefferson County, OK: births 68, ppw_sud 0, bh_clin 0, ob 0, hc 0, need 78.2, cap 16.0, gap 62.2
  largest gap 29119 McDonald County, MO: births 326, ppw_sud 0, bh_clin 9, ob 0, hc 1, need 86.8, cap 27.0, gap 59.8
  smallest gap 36113 Warren County, NY: births 521, ppw_sud 2, bh_clin 347, ob 42, hc 13, need 23.8, cap 93.6, gap -69.8
  smallest gap 08117 Summit County, CO: births 255, ppw_sud 2, bh_clin 156, ob 15, hc 5, need 28.4, cap 94.4, gap -66.0
  smallest gap 32029 Storey County, NV: births 18, ppw_sud 0, bh_clin 21, ob 2, hc 1, need 16.4, cap 81.2, gap -64.8
  smallest gap 34019 Hunterdon County, NJ: births 1064, ppw_sud 1, bh_clin 467, ob 27, hc 1, need 2.8, cap 67.2, gap -64.4
  smallest gap 31119 Madison County, NE: births 465, ppw_sud 2, bh_clin 246, ob 12, hc 7, need 24.0, cap 87.8, gap -63.8
  random   46093 Meade County, SD: births 290, ppw_sud 0, bh_clin 140, ob 2, hc 1, need 31.4, cap 54.7, gap -23.3
  random   54073 Pleasants County, WV: births 62, ppw_sud 0, bh_clin 2, ob 0, hc 1, need 70.9, cap 36.7, gap 34.2
  random   49015 Emery County, UT: births 118, ppw_sud 1, bh_clin 13, ob 2, hc 2, need 55.8, cap 73.7, gap -17.9
  random   41033 Josephine County, OR: births 803, ppw_sud 1, bh_clin 914, ob 28, hc 2, need 83.2, cap 76.6, gap 6.6
  random   20057 Ford County, KS: births 581, ppw_sud 1, bh_clin 47, ob 4, hc 3, need 39.7, cap 53.7, gap -14.0